

Dulvin Hemsara Bandara

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4th-year Aerospace Engineering and Commerce student at Monash University and current Senior Aerodynamics Engineer at Monash Motorsport, leading the Aerodynamics Cooling Subsection. Specialising in CFD workflow development with ANSYS Fluent, PyFluent & MATLAB on HPC, alongside full-car external and thermal aerodynamics. Seeking a 2027 Industrial Placement.

RELEVANT EXPERIENCE

Senior Aerodynamics Engineer - Monash Motorsport - Formula Student

Clayton, VIC
Jan 2025 - Present

- Co-wrote a Git-powered PyFluent architecture on HPC, reducing overall runtime by 54%.
- Established workflow validation via mesh independence & y+ value studies and turbulence model selections.
- Designed and validated internal and external ducting for accumulator cooling, increasing mass flow rates by 400% and targeting a 65% peak cell temperature margin to rules.
- Automated MATLAB and ParaView post-processing tools integrated in the PyFluent environment for generating force, pressure, & thermal contours for design reviews.
- Manufactured 2025 & 2026 aerodynamics packages, with carbon lay-ups and jigging, producing 70 parts over the course of 2 years.
- Developed onboarding programme for aerodynamics recruits; serving as first point of contact and mentor for all new team members across their first few months in the team.

Australian Centre for Sports Aerodynamics

Wind Tunnel Test Engineer (Monash Motorsport)

South Australia
Feb 2026

- Conducted a week of wind tunnel testing across 2 Monash Motorsport vehicles to validate the current CFD model, investigating turbulence modelling and validating geometric independence across two configurations
- Manufactured a tapped wing set connecting to custom DPMS boxes, developing MATLAB scripts to post-process force data and generate pressure plots for pitot and transverse tube placement analysis

Monash High Powered Rocketry

Dynamics Engineer

Clayton, VIC
Mar 2024 - Mar 2025

- Collaborated in a team of 5 to design, manufacture and launch a Tripoli Level 1 rated rocket achieving an apogee of 2,478 ft (0.88% error against 2,500 ft target)
- Developed and implemented ANSYS Watertight meshing workflow for improved CFD accuracy on HPC

EDUCATION

Monash University

Bachelor of Aerospace Engineering (Honours) & Bachelor of Commerce majoring in Finance

Clayton, VIC
Dec 2028

LEADERSHIP & ACTIVITIES

Motorsport Australia

Race Marshal

2024 - Present

- Active race marshal across multiple events, including V8 Supercars, Shannons Speed Series, and the 2026 Australian Grand Prix

Monash Aerospace & Mechanical Engineering Club

Communications & Sponsorship Manager

Aug 2023 - March 2026

- Coordinated with sponsors and leading aerospace and mechanical companies to organise Industry Nights and nurture club relations

John Monash Science School

Co-Curricular Facilitator - Introduction to Aerospace Engineering

Clayton, VIC
Early 2026 – Present

- Developing and delivering a basic Aerospace curriculum to Grade 10 and 11 students.

SKILLS & INTERESTS

CFD & Simulation: ANSYS Fluent, PyFluent, OpenFOAM (Basic), SU2 (Basic), Paraview

CAD: Siemens NX, SolidWorks, Fusion 360

Programming: Python, MATLAB, R, LaTeX, Git, HPC

Languages: English (native), Sinhalese (native), French (Conversational), Italian (Conversational)

Interests: Formula 1, Karting, Amateur Cricket, University Running and Volleyball Clubs